

Stage 4

Volume

Book 1

Version: 260912
Feedback:
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Learning Intention**Error! Bookmark not defined.**

Syllabus Content

MA4-VOL-C-01 applies knowledge of volume and capacity to solve problems involving right prisms and cylinders

Describe the different views of prisms and solids that have been formed from prism combinations

- Represent prisms from different views in 2 dimensions, including top, side, front and back views
- Describe and illustrate solids formed from prism combinations from different views in 2 dimensions, including top, side, front and back views
- Identify and illustrate the cross-sections of different prisms
- Examine the idea that prisms have a uniform cross-section that is equal to the base area
- Determine if a particular solid has a uniform cross-section

Develop and apply the formula to find the volume of a prism to solve problems

- Develop the formula for the volume of a prism: $V = \text{base area} \times \text{height}$, leading to $V = Ah$
- Apply the formula for the volume of a prism to prisms with uniform cross-sections to solve problems

Develop the formula for finding the volume of a cylinder and apply the formula to solve problems

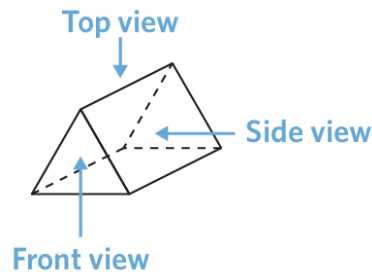
- Develop and apply the formula to solve problems involving the volume of cylinders: $V = \pi r^2 h$, where r is the length of the radius of the base and h is the perpendicular height

Choose appropriate units of measurement for volume and capacity and convert between units

- Recognise that 1000 L is equal to 1 kilolitre (kL) and use the abbreviation
- Recognise that 1000 kL is equal to 1 megalitre (ML) and use the abbreviation
- Choose an appropriate unit to measure the volume or capacity of different objects and justify the choice
- Convert between metric units of volume and capacity
($1 \text{ cm}^3 = 1000 \text{ mm}^3$, $1 \text{ cm}^3 = 1 \text{ mL}$, $1 \text{ m}^3 = 1000 \text{ L} = 1 \text{ kL}$)
- Solve practical problems involving the volume and capacity of right prisms and cylinders

Drawing Prisms

 **Example** Draw the top, side, front, and back view of a simple prism



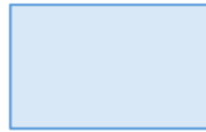
Front



Back



Side



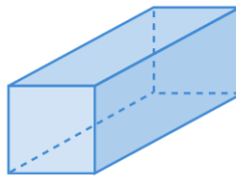
Top



 **Guided Practice**

Sketch the front, back, side, and top views of these prisms.

a



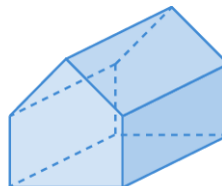
Front

Back

Side

Top

b



Front

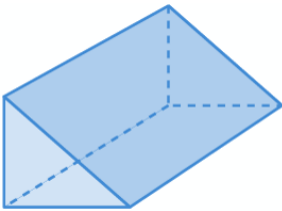
Back

Side

Top

1 Sketch the front, back, side, and top views of these prisms.

a



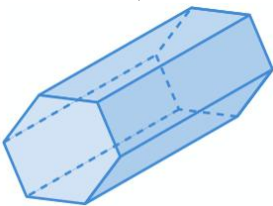
Front

Back

Side

Top

b



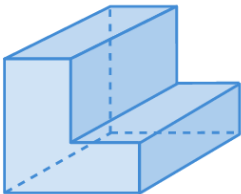
Front

Back

Side

Top

c



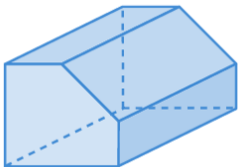
Front

Back

Side

Top

d



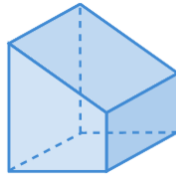
Front

Back

Side

Top

e



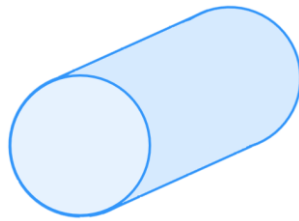
Front

Back

Side

Top

f



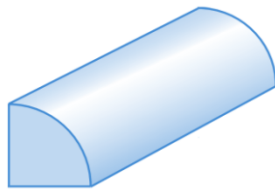
Front

Back

Side

Top

g



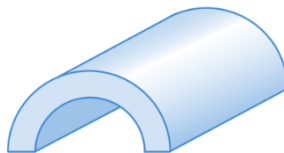
Front

Back

Side

Top

h



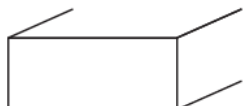
Front

Back

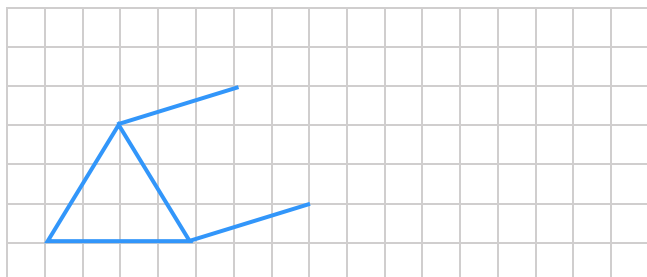
Side

Top

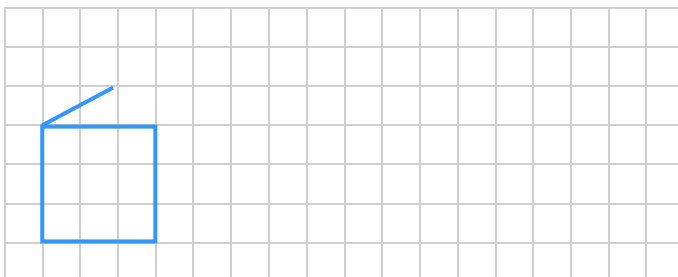
- 2 Consider the example of how to sketch a rectangular prism and sketch these prisms.



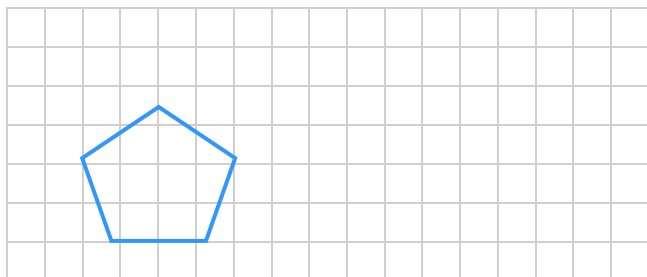
a Triangular Prism



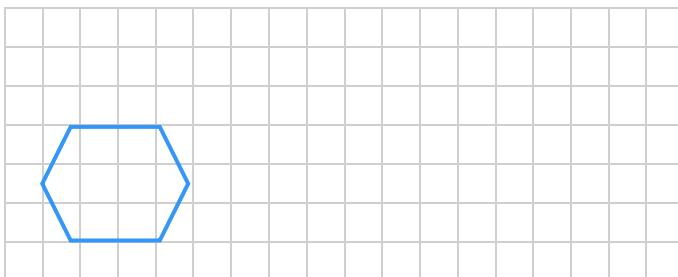
b Cube



c Pentagonal Prism

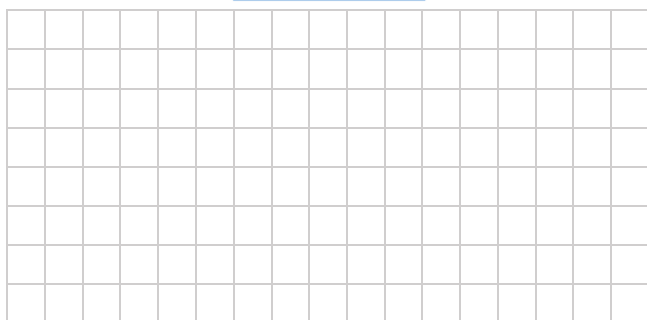


d Hexagonal Prism



- 3 Sketch a prism with the following front and back faces:

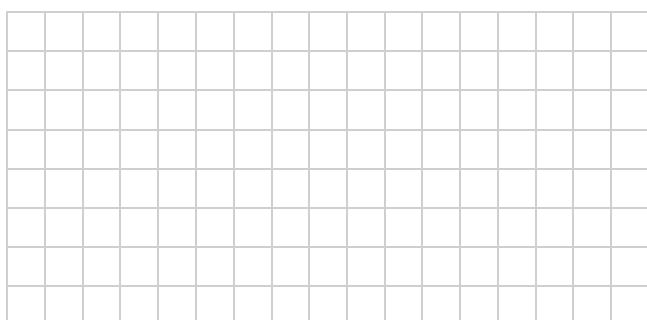
a



b



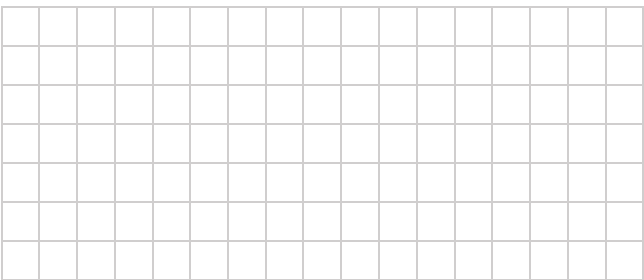
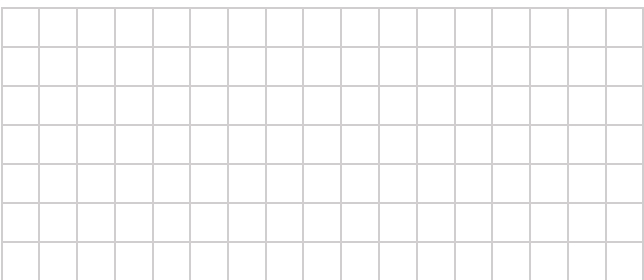
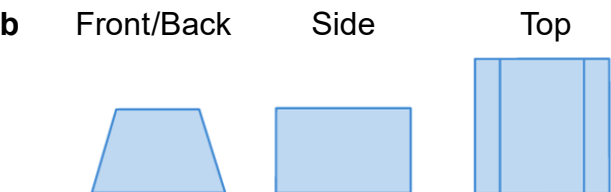
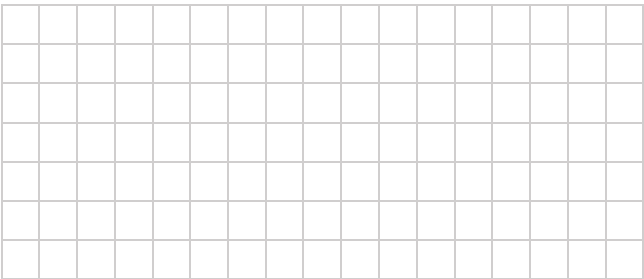
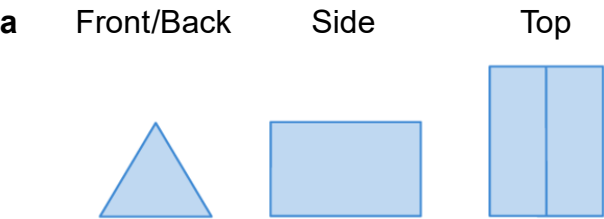
c



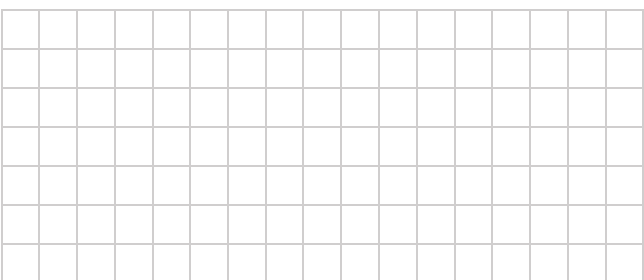
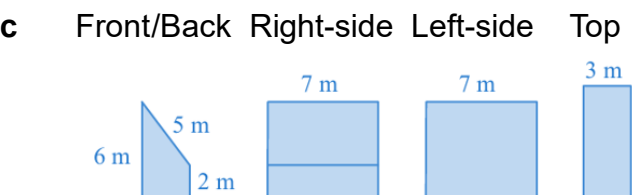
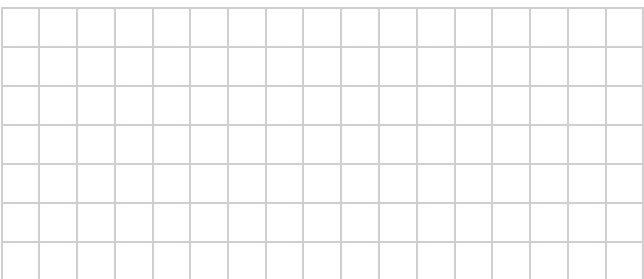
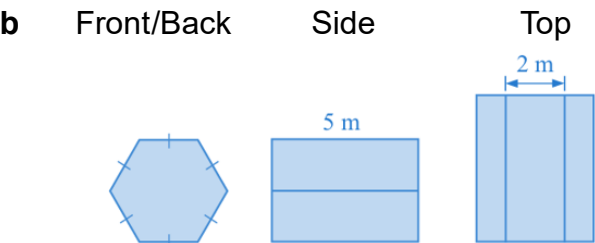
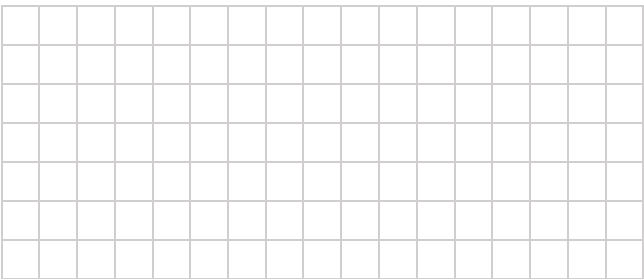
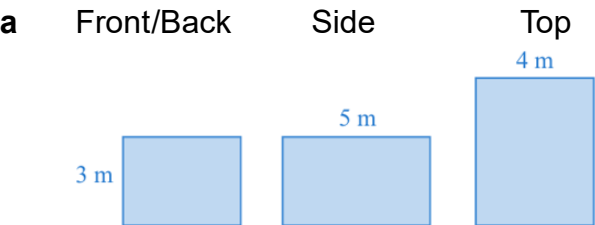
d



4 Sketch these prisms using the given faces.



5 Sketch these prisms using the given faces and label the dimensions.

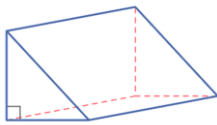


Cross-Section of a Prism

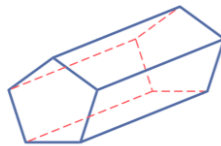
i Prisms

- A **prism** is a type of 3D solid with identical opposite end faces connected by rectangular sides.
- The **base of a prism** is the identical opposite end faces.
- The **height of a prism** is the perpendicular distance between the bases.
- We name prisms based on the shape of their base.

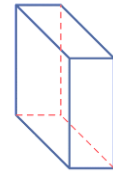
Triangular Prism



Pentagonal Prism

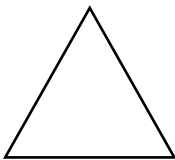


Rectangular Prism



Q Identify prisms

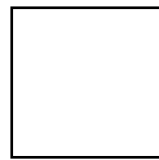
a



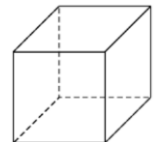
b



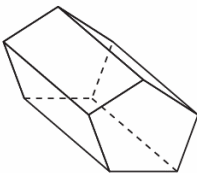
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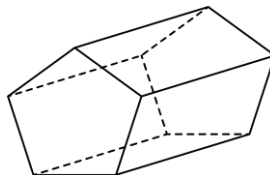
d



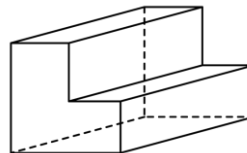
e



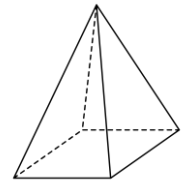
f



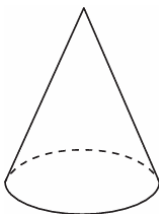
g



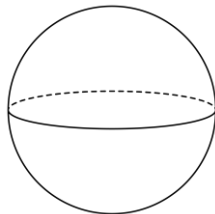
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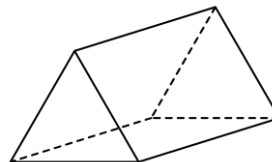
i



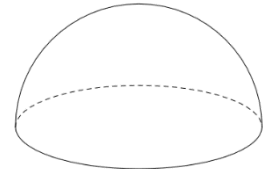
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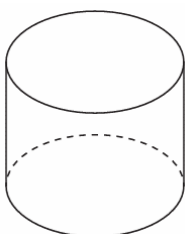
k



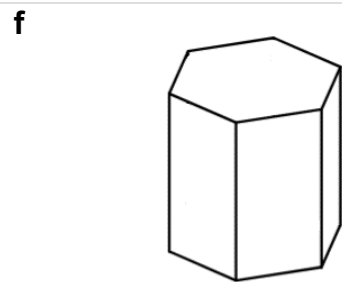
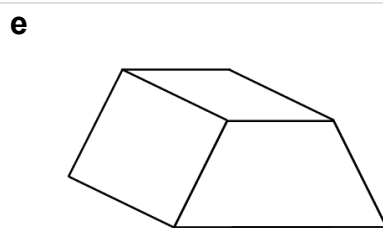
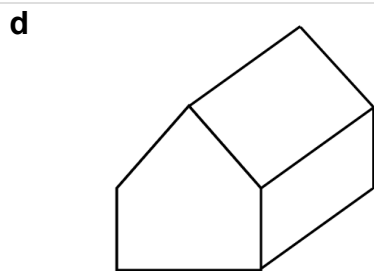
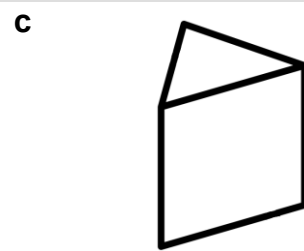
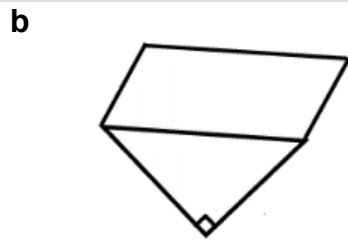
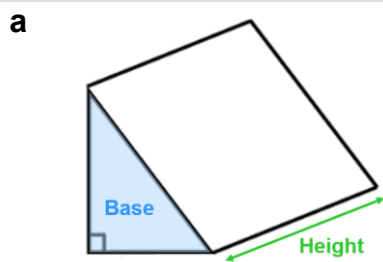
l



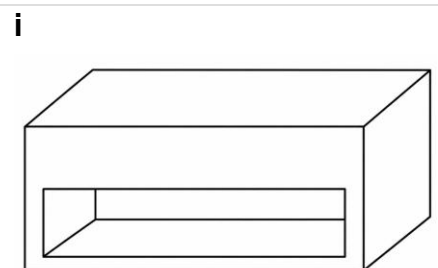
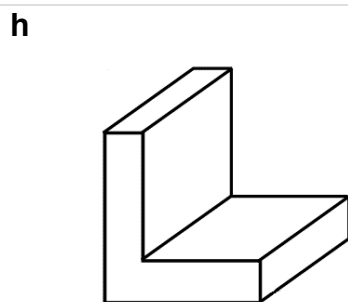
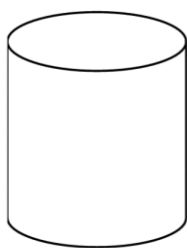
- m** The base of a prism must be a polygon; a polygon has straight sides. Explain why a cylinder is **not** a prism.



🔍 **Identify** base and height of solids with uniform cross-section

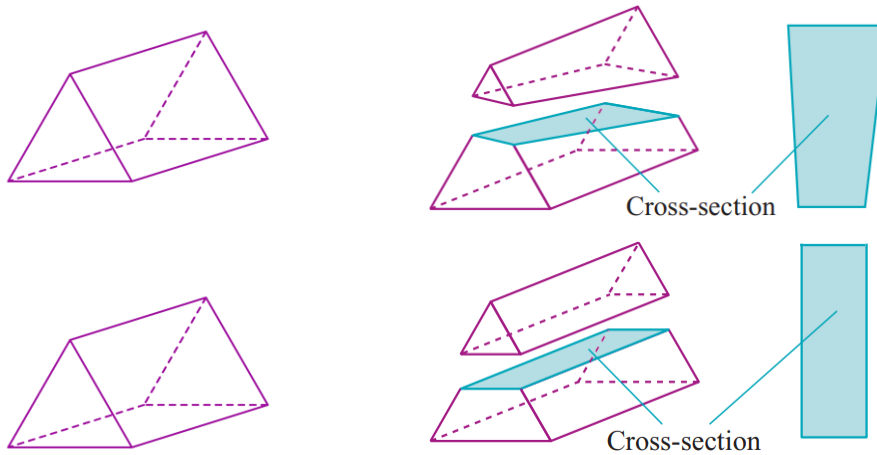


g *cylinders are not prisms.*

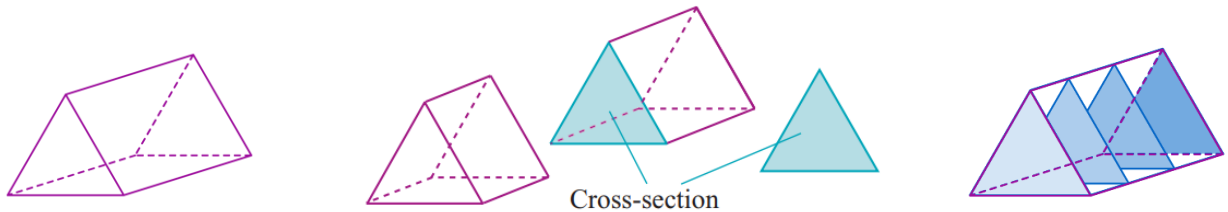


i Cross-Section of a Prism

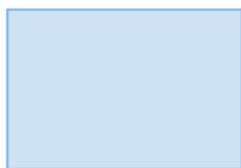
- A **cross-section** shows the shape formed when we 'slice' a solid.



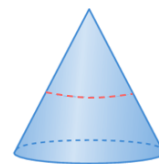
- A prism has **uniform cross section**; any cross section **parallel** to the base is identical to the base.



🔧 Example Draw the cross section of a solid

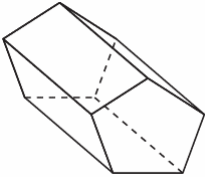
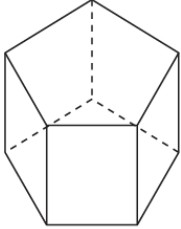
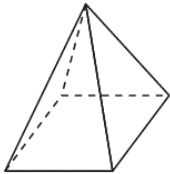
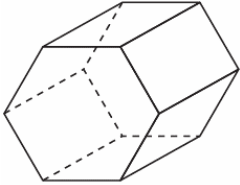
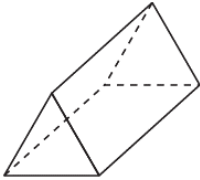
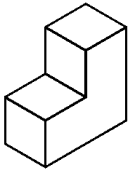


Explain why this is a prism.



Explain why this is not a prism.

Guided Practice

	Solid	Cross-section parallel to base	Prism?
a			
b			
c			
d			
e			
f			
g			

Key Ideas

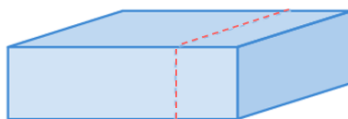
1. A prism is a-dimensional solid with identical faces connected by sides.
2. A - is the shape formed from “.....” the object.
3. A prism has cross-section; slices to the base are identical to the base.

1 Draw the cross-section along the dotted line.

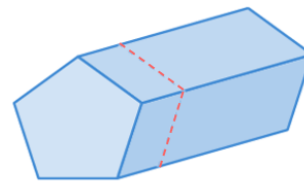
a



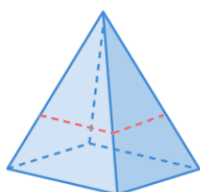
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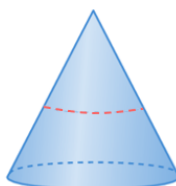
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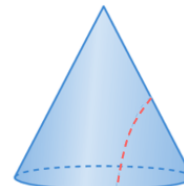
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e

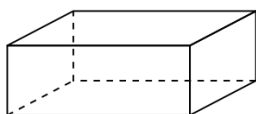


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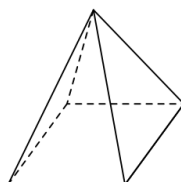
2 Decide whether these solids have uniform cross-section parallel to the base.

a



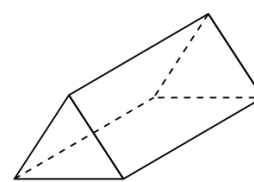
yes

b



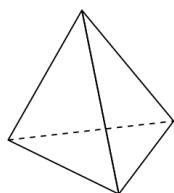
no

c



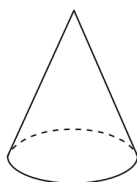
yes

d



no

e



no

f



yes

g



no

h



no

i



yes

j



yes

k



yes

l



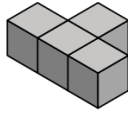
no

Volume of a Prism (Base Area Given)

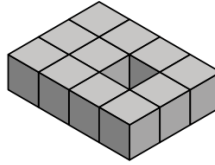
Review

- ☐ Calculate volume by counting cubes. Each cube is 1 cm^3 .

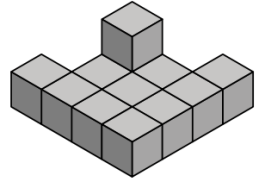
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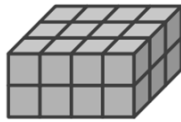
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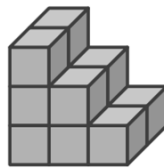
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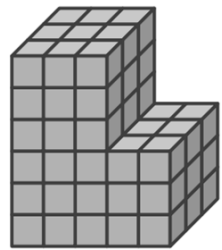
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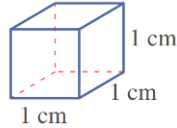
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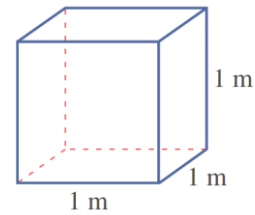
i Volume

- **Volume** is the amount of space taken up by a 3-dimensional object.
- Volume is measured in **cubic units**.

cubic centimetre (cm^3)



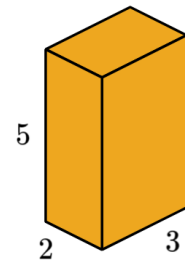
cubic metre (m^3)



Q Volume of a Prism

Suppose we want to find how many unit cubes can fit in this box.

The box is a special type of 3D solid called a , because it has -

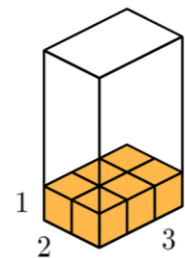


The number of unit cubes that can fit in the box is called the

We will begin by considering the bottom layer, or base, of the shape.

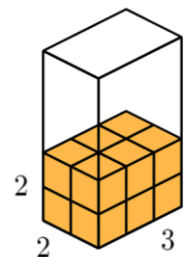
The number of cubes that can fit in the bottom layer is

..... $\times$ =



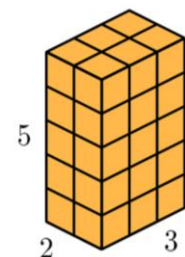
If there are two layers, then the number of cubes is:

..... $\times$ $\times$ =



So, if there are 5 layers, then the number of cubes is:

..... $\times$ $\times$ =



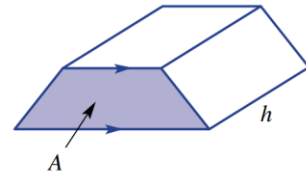
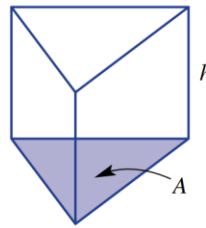
The volume of any prism can be found using the formula:

Volume = Area of the b..... $\times$ number of l.....

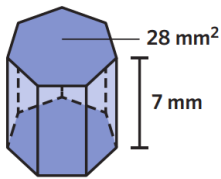
<https://www.geogebra.org/m/rqnd3khf>

i Volume of a Prism

Base Area
↓
 $V = Ah$
↑ ↑
Volume Height



Example Calculate volume of a prism (base area given)

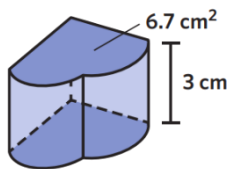


$$\begin{aligned} V &= Ah \\ &= 28 \text{ mm}^2 \times 7 \text{ mm} \\ &= 196 \text{ mm}^3 \end{aligned}$$

Guided Practice

Calculate the volume of these prisms

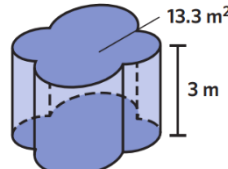
a



$$\begin{aligned} V &= Ah \\ &= \dots\dots\dots \\ &= \dots\dots\dots \end{aligned}$$

20.1 cm³

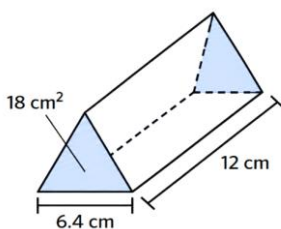
b



$$\begin{aligned} V &= Ah \\ &= \dots\dots\dots \\ &= \dots\dots\dots \end{aligned}$$

39.9 m³

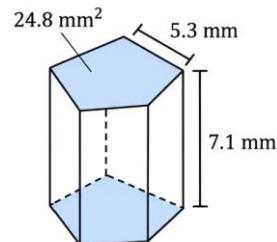
c



$$\begin{aligned} V &= Ah \\ &= \dots\dots\dots \\ &= \dots\dots\dots \end{aligned}$$

216 cm³

d



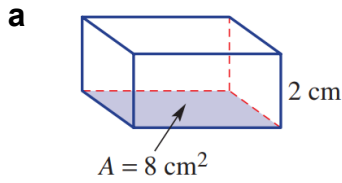
$$\begin{aligned} V &= Ah \\ &= \dots\dots\dots \\ &= \dots\dots\dots \end{aligned}$$

176.08 m³

Key Ideas

1. Volume is how much a 3D solid takes up.
2. Volume is measured using units.
3. Calculate the volume of a prism by multiplying the base by the

1 The area of the base is given. Calculate the volume of these prisms.

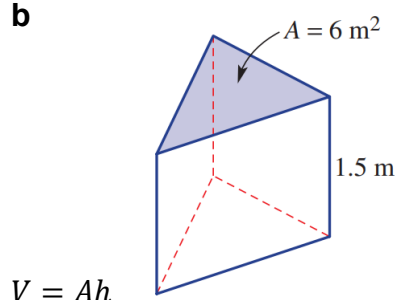


$$V = Ah$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

16 cm³

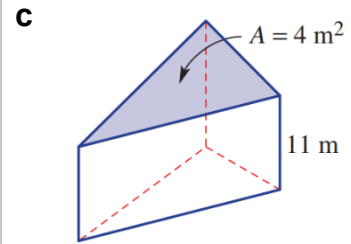


$$V = Ah$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

9 m³

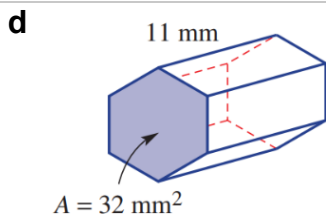


$$V = Ah$$

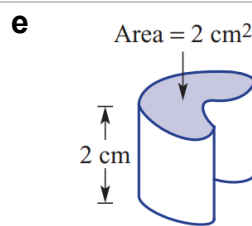
$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

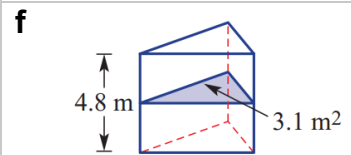
44 m³



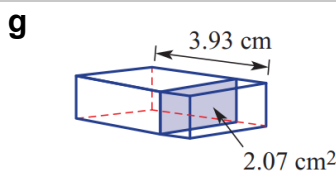
352 mm³



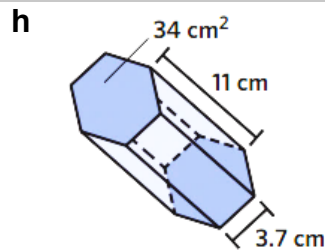
4 cm³



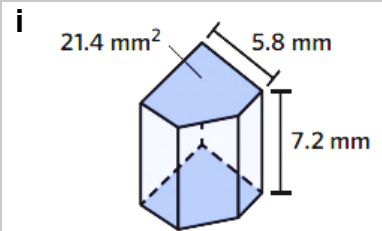
14.88 m³



8.1351 cm³

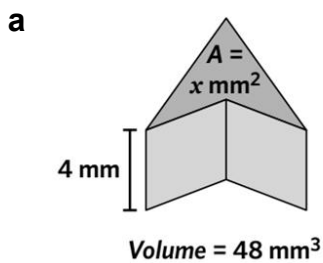


374 cm³



154.08 mm³

2 Find the missing value for each pronumeral.

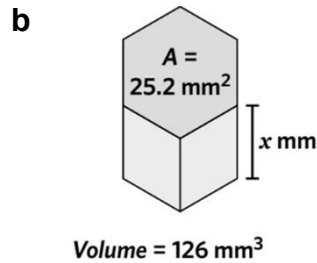


$$V = Ah$$

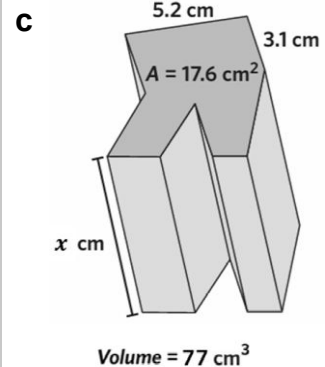
$$48 = 4x$$

$$\dots\dots = x$$

12 mm



5 mm



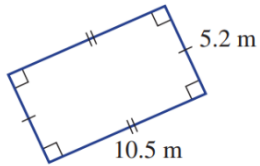
0.4375 cm³

Volume of a Prism

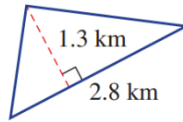
Review

☐ Calculate area of simple shapes

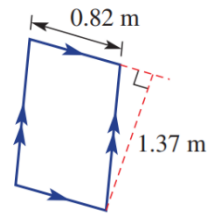
a



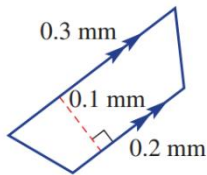
b



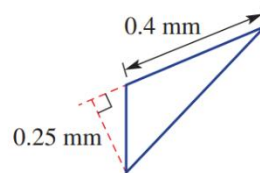
c



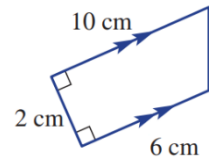
d



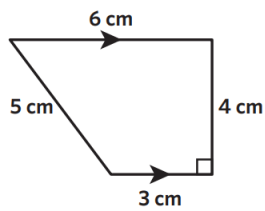
e



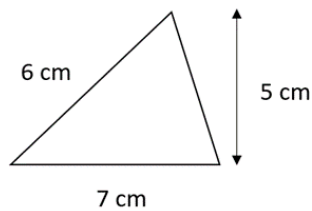
f



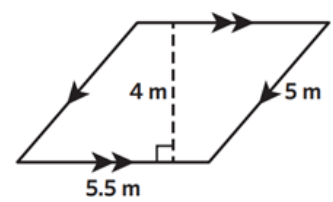
g



h



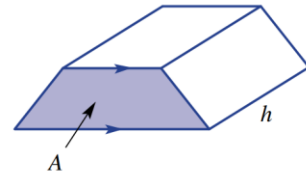
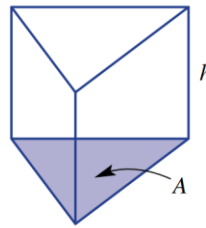
i



i Volume of a Prism

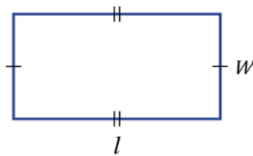
$$V = Ah$$

Base Area
 ↓
 Volume Height



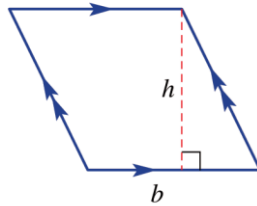
Rectangle

$$A = lw$$



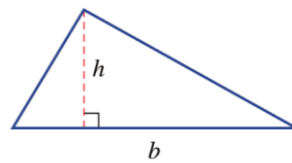
Parallelogram

$$A = bh$$



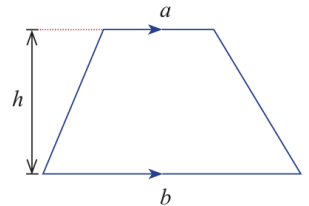
Triangle

$$A = \frac{bh}{2}$$

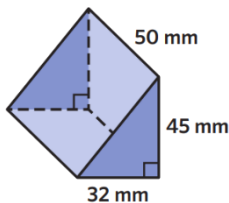


Trapezium

$$A = \frac{h(a+b)}{2}$$



Example Calculate volume of a prism



$$\begin{aligned}
 V &= Ah \\
 &= \left(\frac{45 \times 32}{2} \right) \times 50 \\
 &= 36\,000 \text{ mm}^3
 \end{aligned}$$

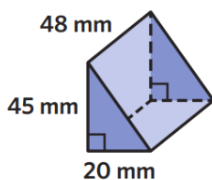
The base of this prism is a because that is the face with uniform -

The height of this prism is, it is to the height of the triangle.

Guided Practice

Calculate the volume of these prisms.

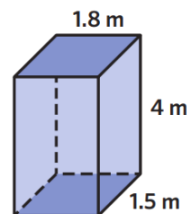
a



$$\begin{aligned}
 V &= Ah \\
 &= \left(\frac{\times}{2} \right) \times \dots \\
 &= \dots
 \end{aligned}$$

21600 mm³

b

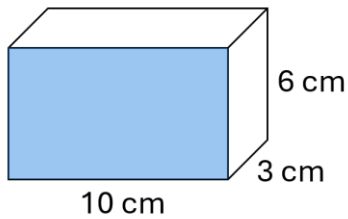
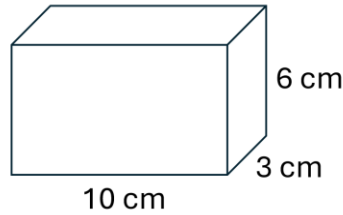


$$\begin{aligned}
 V &= Ah \\
 &= (\dots \times \dots) \times \dots \\
 &= \dots
 \end{aligned}$$

10.8 m³

🔍 Volume of a Rectangular Prism

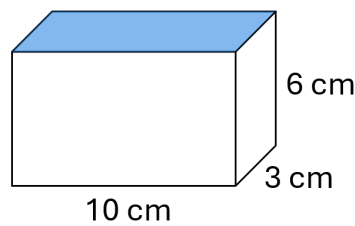
A rectangular prism is special because you can choose any of the three faces as the base of the prism.



$$V = Ah$$

$$= (\dots \times \dots) \times \dots$$

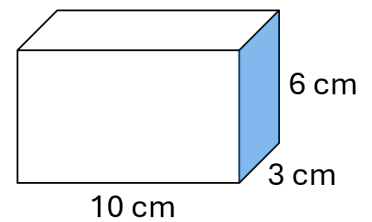
$$= \dots \text{ cm}^2$$



$$V = Ah$$

$$= (\dots \times \dots) \times \dots$$

$$= \dots \text{ cm}^2$$



$$V = Ah$$

$$= (\dots \times \dots) \times \dots$$

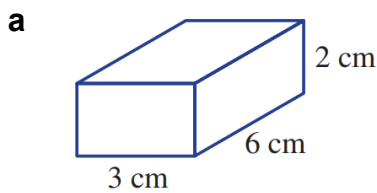
$$= \dots \text{ cm}^2$$

So, for a rectangular prism, you can the volume is the product of the three dimensions:

$$\begin{aligned} V &= \text{length} \times \text{width} \times \text{height} \\ &= lwh \end{aligned}$$

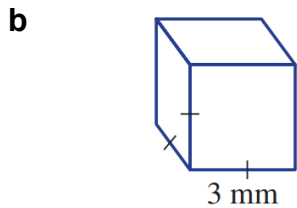
For any other type of prism, there is only one way to choose the base area, so there isn't a nice shortcut.

1 Calculate the volumes of these solids.



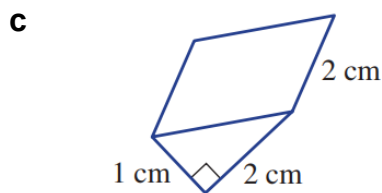
$V = Ah$
=
=

36 cm³



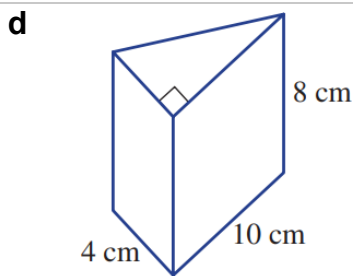
$V = Ah$
=
=

27 cm³

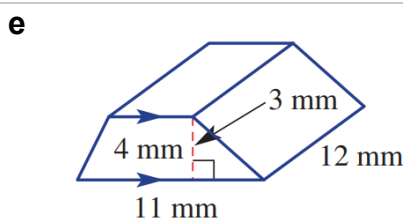


$V = Ah$
=
=

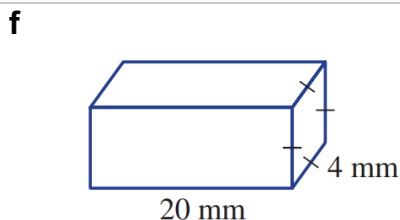
2 cm³



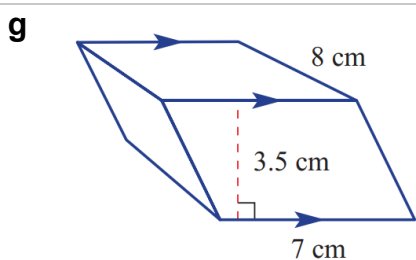
160 cm³



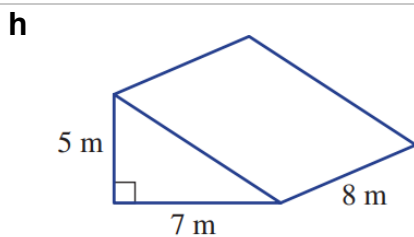
270 mm³



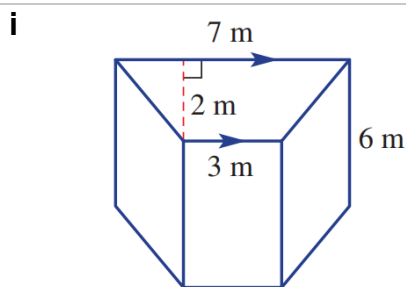
320 mm³



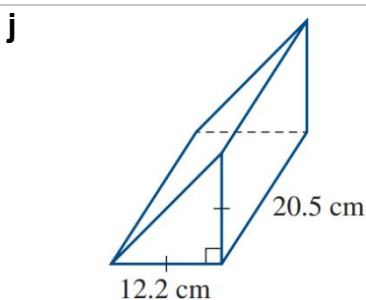
196 cm³



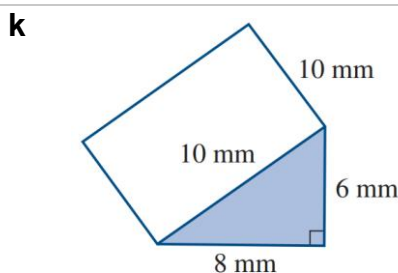
140 m³



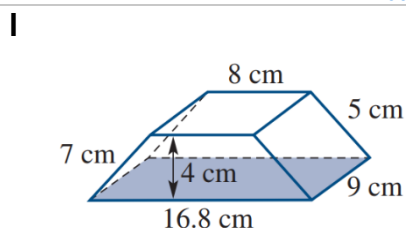
60 m³



1525.61 cm³

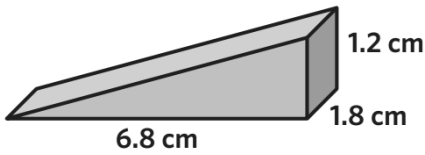


240 mm³



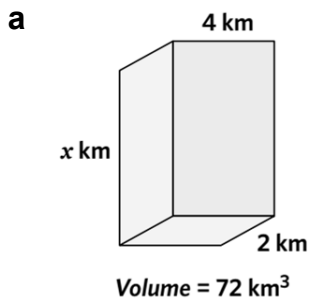
446.4 cm³

- 2 Charya is going to make 100 doorstops for his school with recycled plastic. The doorstops will be in the shape of a triangular prism with these dimensions. What volume of plastic will he need to make all 100 door stops?

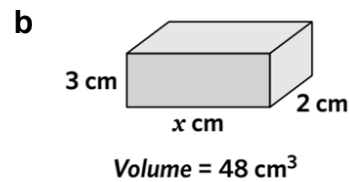


734.4 cm³

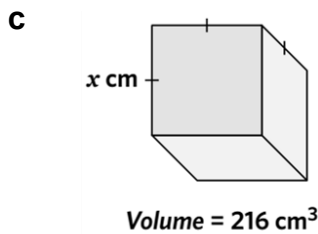
- 3 Find the missing value for each pronumeral.



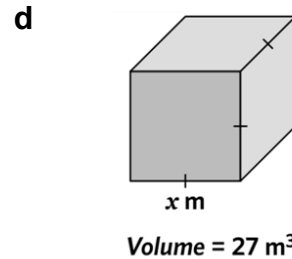
9 km



8 cm

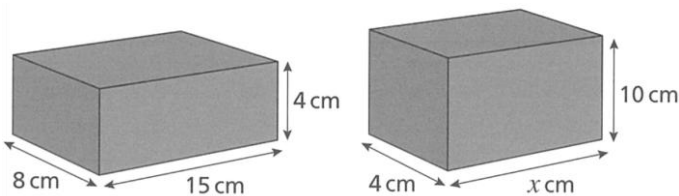


6 cm



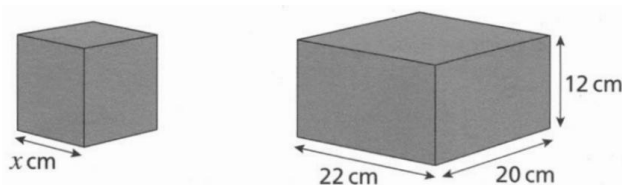
3 m

- 4 Both rectangular prisms have the same volume. Work out the value of x .



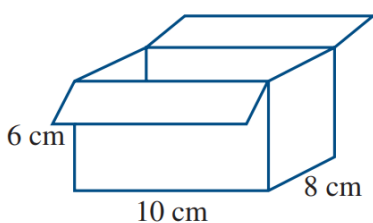
12 cm

- 5 The volume of the rectangular prism is double the volume of the cube. Work out the value of x to 2 d.p.



13.82 cm

- 6 A vase with a volume of 200 cm³ is packed into the cardboard box shown below. The space around the vase is filled with packing foam. What is the volume of the foam?



280 cm³

7

A cube measures 5 cm by 5 cm by 5 cm.

a Work out the volume of the cube.

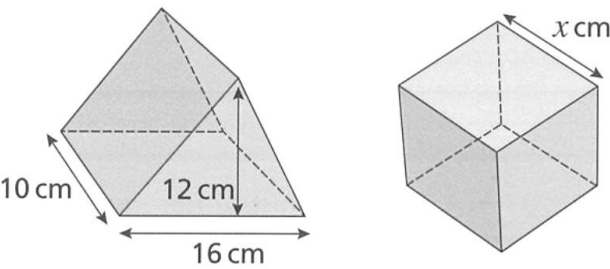
125 cm³

b The length of all the sides is increased by 10%.
Calculate the percentage increase in the volume of the cube.

$$1.1^3 = 1.331$$

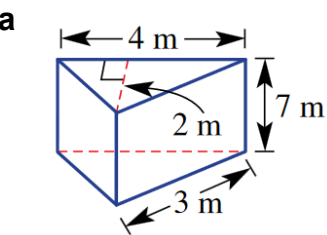
33.1%

8 The cube and triangular prism have the same volume. Work out the value of x to 2 d.p.

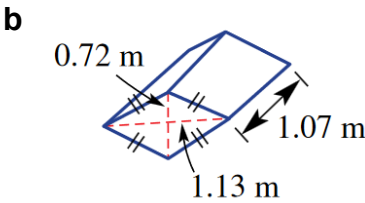


9.86 cm

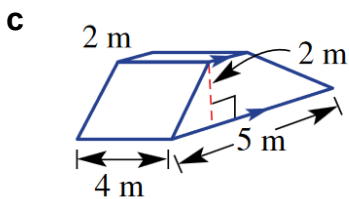
9 Find the volume of each of the following solids.



28 m³

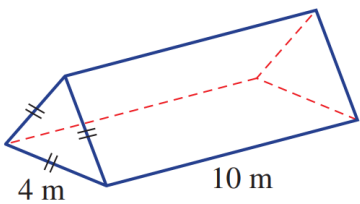


0.435276 m³



28 m³

10 Determine the volume of this equilateral triangular prism. Answer to 1 d.p.
(hint: you need to use Pythagoras' theorem)

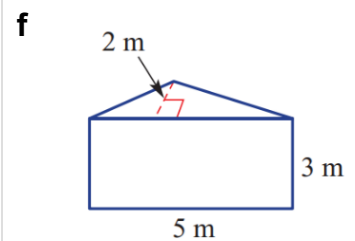
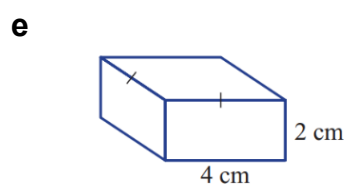
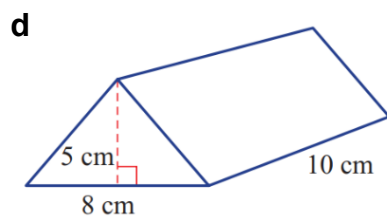
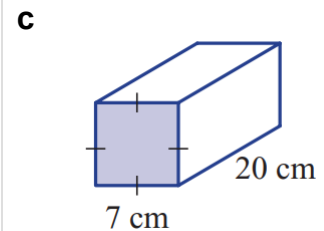
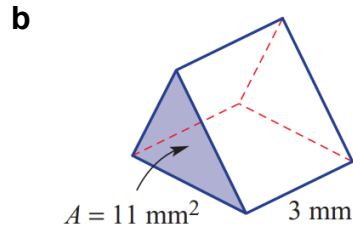
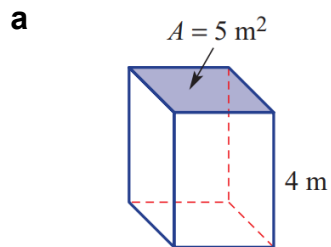


69.3 m³

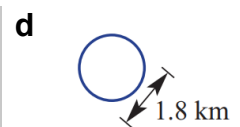
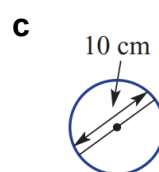
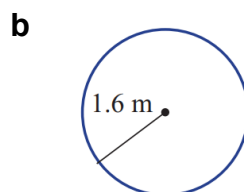
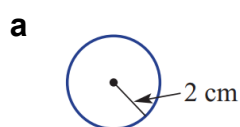
Volume of a Cylinder

Review

- ☐ Calculate the volume of these prisms.



- ☐ Calculate area of these circles.
Give your answer in terms of π and rounded to two decimal places.



$$A = \pi r^2$$

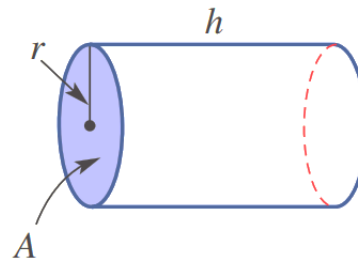
=

=

i Volume of a Cylinder

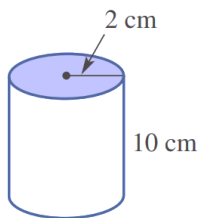
- A **cylinder** is a 3D solid with uniform cross-section of a circle.
- The volume of a cylinder is calculated using $V = Ah$, where $A = \pi r^2$

$$\begin{array}{c}
 \text{Radius} \\
 \downarrow \\
 V = \pi r^2 h \\
 \begin{array}{cc}
 \uparrow & \uparrow \\
 \text{Volume} & \text{Height}
 \end{array}
 \end{array}$$

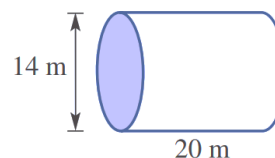


Example Calculate volume of a cylinder

Calculate the volume of these cylinders, round your answer to 2 d.p.



$$\begin{aligned}
 V &= \pi r^2 h \\
 &= \pi \times 2^2 \times 10 \\
 &= 125.66 \text{ cm}^3
 \end{aligned}$$

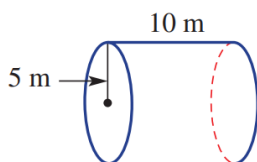


$$\begin{aligned}
 V &= \pi r^2 h \\
 &= \pi \times 7^2 \times 20 \\
 &= 3078.76 \text{ m}^3
 \end{aligned}$$

Guided Practice

Calculate the volume of these cylinders. Round your answer to 2 decimal places.

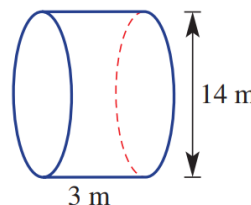
a



$$\begin{aligned}
 V &= Ah \\
 &= \dots\dots\dots \\
 &= \dots\dots\dots
 \end{aligned}$$

785.40 m³

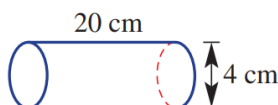
b



$$\begin{aligned}
 V &= Ah \\
 &= \dots\dots\dots \\
 &= \dots\dots\dots
 \end{aligned}$$

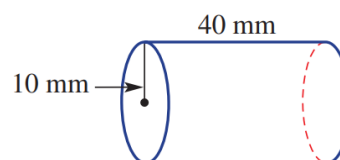
461.81 m³

c



251.33 cm³

d



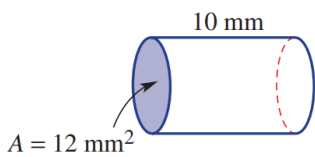
12566.37 mm³

Key Ideas

- A cylinder is similar to a prism because it has
- The volume of any solid with uniform cross section is $V = \dots\dots\dots$
- The volume of a cylinder is $V = \pi r^2 h$ because the area of the base is

1 The area of the base is given. Calculate the volume of these cylinders.

a



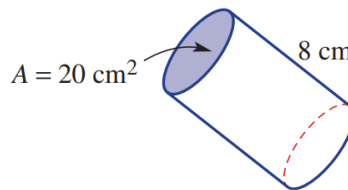
$$V = Ah$$

$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

120 mm³

b



$$V = \dots\dots\dots$$

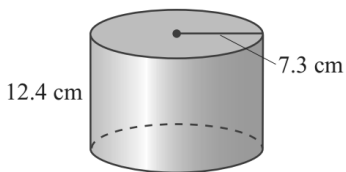
$$= \dots\dots\dots$$

$$= \dots\dots\dots$$

160 cm³

2 Complete to calculate the volume of the cylinder to the nearest cubic cm.

a



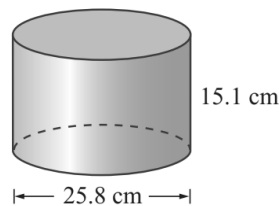
$$V = \pi r^2 h$$

$$= \pi \times \dots^2 \times \dots$$

$$= \dots\dots\dots \text{cm}^3$$

2076 cm³

b



$$V = \pi r^2 h$$

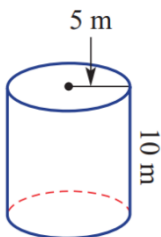
$$= \pi \times \dots^2 \times \dots$$

$$= \dots\dots\dots \text{cm}^3$$

7894 cm³

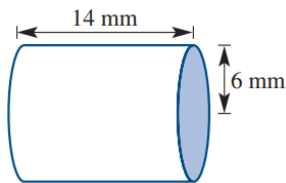
3 Calculate the volume of these cylinders. Give your answer in exact form in terms of π .

a



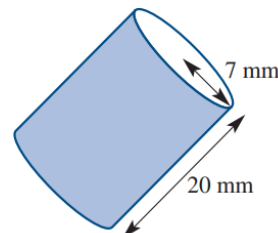
250 π m³

b



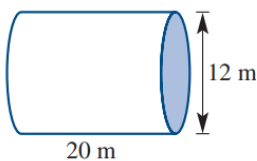
504 π mm³

c



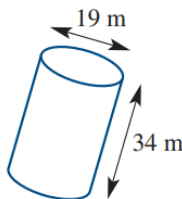
980 π mm³

d



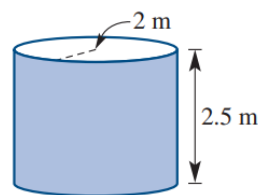
720 π m³

e



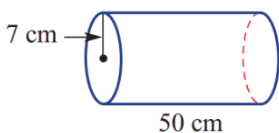
3068.5 π m³

f



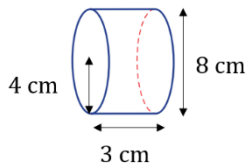
10 π m³

g



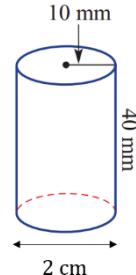
2450 π cm³

h



48 π cm³

i

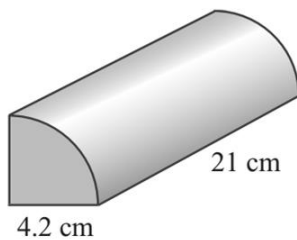


4000 π mm³

- 4 A cylindrical tank has a diameter of 3 m and height 2 m.
Calculate the volume in cubic metres (m^3), answer to 1 d.p.

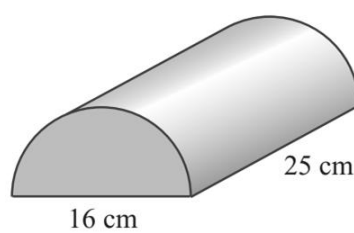
- 5 Calculate the volume of these fractions of a cylinder. Answer to 2 d.p.

a



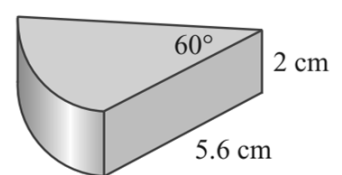
290.94 cm^3

b



2513.27 cm^3

c



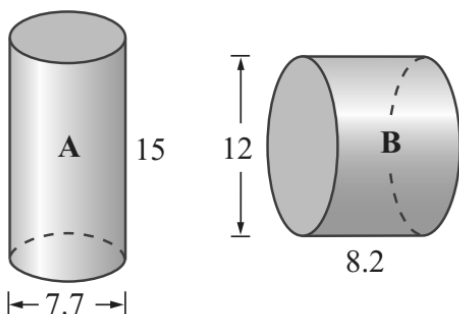
32.84 cm^3

14.1 cm^3

- 6 A cylindrical tin of beans has radius 4 cm and a height of 10 cm.
The can contains 400 cm^3 of beans. How much empty space is there inside the tin?

102.65 cm^3

- 7 Determine which of these prisms has the largest volume.



Cylinder A: 698.5

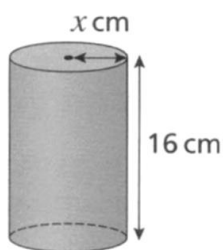
Cylinder B: 927.4

Cylinder B is larger.

- 8 Determine the unknown dimension of the cylinder.

a

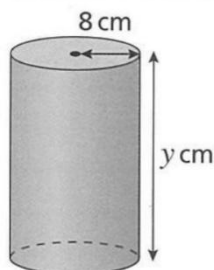
$$V = 615.8 \text{ cm}^3$$



$$x = 3.5 \text{ cm}$$

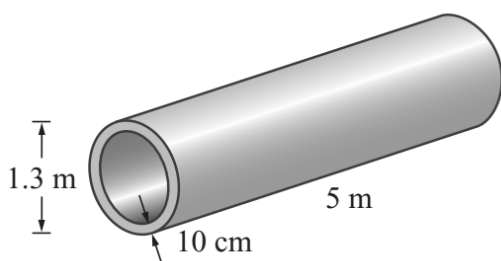
b

$$\text{Volume} = 2513 \text{ cm}^3$$



$$y = 12.5 \text{ cm}$$

- 9 Concrete sewer pipes are 5 m long with an outer diameter of 1.3 m. The walls of the pipes are 10 cm thick. What volume of concrete is required to make each pipe? Express your answer in m^3 correct to 2 decimal places.



$$\text{Large cylinder} - \text{small cylinder} = \pi(0.65)^2(5) - \pi(0.55)^2(5) = 1.88 \text{ m}^3$$

- 10 A cylindrical wheat hopper is 8 m in diameter and 10 m high.

a Calculate its volume correct to two decimal places.

$$V = \pi(4)^2(10) = 502.65 \text{ m}^3$$

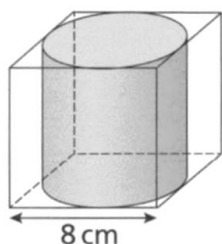
b Grain pours out at $48\,000 \text{ cm}^3$ per second. How long would it take a full hopper to empty? Answer correct to the nearest minute. $1\,000\,000 \text{ cm}^3 = 1 \text{ m}^3$

$$175 \text{ minutes.}$$

- 11 A cube has every single one of its sides doubled. How many times is the new volume compared to the original?

$$8 \text{ times}$$

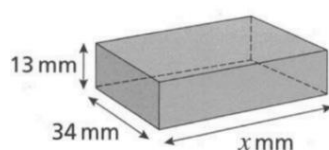
- 12 Express the volume of the cylinder as a percentage of the volume of the cube. Answer to 1 d.p.



$$\frac{V_{\text{Cyl}}}{V_{\text{cube}}} \times 100 = \frac{\pi(4)^2(8)}{8^3} \times 100 = 78.5\%$$

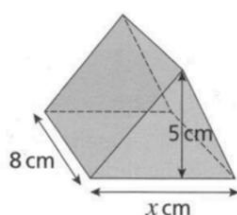
- 13 Find the missing length in each of these shapes.

a $V = 22\,100 \text{ mm}^3$



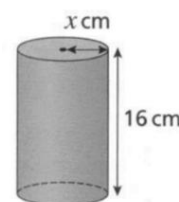
50 mm

b $V = 180 \text{ cm}^3$



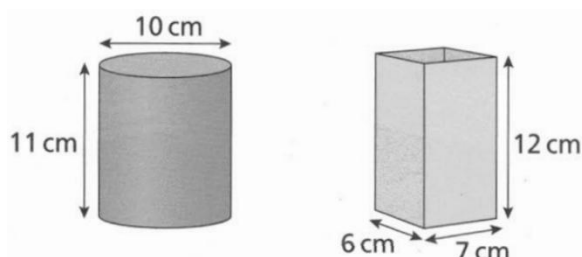
9 cm

c $V = 615.8 \text{ cm}^3$



3.5 cm

- 14 Marta fills the cylindrical container with water. She then pours the water into the rectangular prism-shaped container. Will all of the water fit into the rectangular prism? Justify your answer with mathematical reasoning.

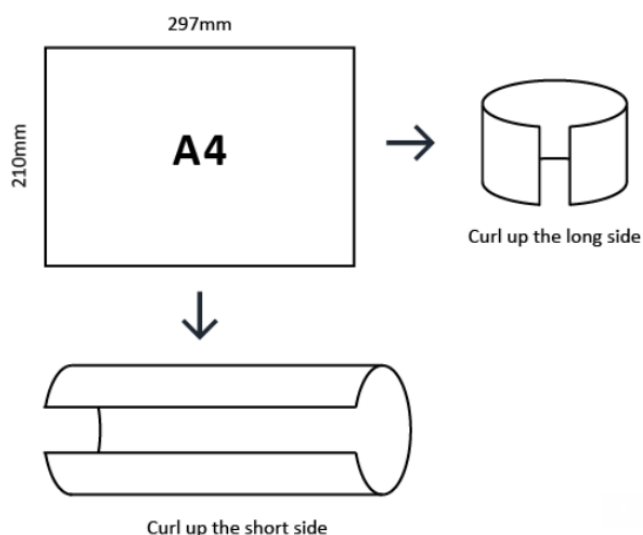


Volume of cylinder: 863.94 cm^3

Volume of rectangular prism: 504 cm^3

No, the volume of the prism is less than the cylinder.

- 15 Which cylinder would have a greater volume? Justify your answer with mathematical reasoning.



Short side: $C = 2\pi r = 210, r = \frac{210}{2\pi}$

$$V_{\text{curl short side}} = \pi \left(\frac{210}{2\pi} \right)^2 (297) \approx 1042282 \text{ cm}^3$$

Long side: $C = 2\pi r = 297, r = \frac{297}{2\pi}$

$$V_{\text{curl long side}} = \pi \left(\frac{297}{2\pi} \right)^2 (210) \approx 1474084 \text{ cm}^3$$

$\therefore$ cylinder curled up long side has larger volume.

Converting Units of Volume

Review

☐ Convert these units of length.

a 5 cm to mm

b 2 m to cm

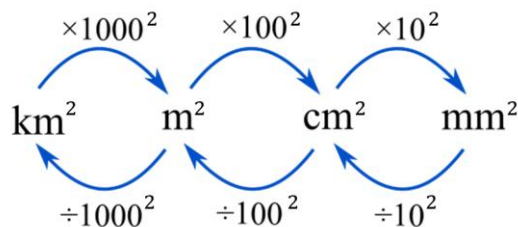
c 3.5 km to m

d 2.1 m to mm

e 400 mm to m

f 500 cm to km

☐ Convert these units of area



a 5 cm^2 to mm^2

b 2 m^2 to cm^2

c 3.5 km^2 to ha

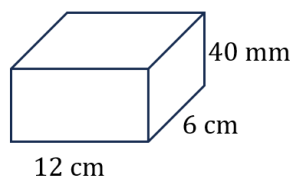
d 2.1 m^2 to mm^2

e 400 mm^2 to m^2

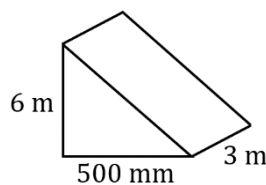
f 500 cm^2 to ha

☐ Calculate the volume of these solids. Answer in the smaller unit.

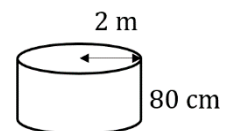
a



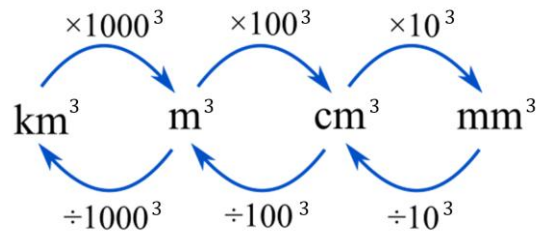
b



c

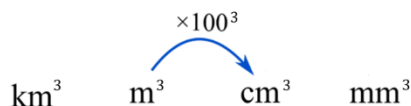


① Converting Units of Volume



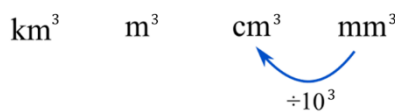
🔧 Example Convert units of volume

20 m³ to cm³



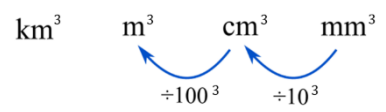
$$20 \text{ m}^3 \times 100^3 \\ = 20\,000\,000 \text{ cm}^3$$

5000 mm³ to cm³



$$5000 \text{ mm}^3 \div 10^3 \\ = 5 \text{ cm}^3$$

150 000 mm³ to m³



$$150\,000 \text{ mm}^3 \div 10^3 \div 100^3 \\ = 0.00015 \text{ m}^3$$

✎ Guided Practice

Convert these units of volume.

a 5 cm³ to mm³

b 2 m³ to cm³

c 40 000 mm³ to m³

d 0.15 m³ to mm³

5000 mm³

e 35 000 m³ to km³

2 000 000 cm³

f 5 500 000 cm³ to m³

0.000 04 m³

150 000 000 mm³

0.000 035 km³

5.5 m³

💡 Key Ideas

- Volume is measured using units;
5 m³ means the volume that 5 cubes of side length would cover.
- To convert between metres and centimetres, the conversion factor is
- To convert between square metres and square centimetres, the conversion factor is
- To convert between cubic metres and cubic centimetres, the conversion factor is

1 Express these volumes in cm^3

a 2.5 m^3

$2\,500\,000 \text{ cm}^3$

b $\frac{3}{4} \text{ m}^3$

$750\,000 \text{ cm}^3$

c 5000 mm^3

5 cm^3

d 100 mm^3

0.1 cm^3

e 62 m^3

$62\,000\,000 \text{ cm}^3$

2 Express these volumes in m^3

a $2\,000\,000 \text{ cm}^3$

2 m^3

b 50000 cm^3

0.05 m^3

c $850\,000 \text{ cm}^3$

0.85 m^3

3 Convert these units of volume.

a 8 cm^3 to mm^3

8000 mm^3

b 3000 mm^3 to cm^3

3 cm^3

c $90\,000\,000 \text{ cm}^3$ to m^3

90 m^3

d 18 m^3 to cm^3

$18\,000\,000 \text{ cm}^3$

e 30 mm^3 to cm^3

0.03 cm^3

f 0.02 m^3 to cm^3

$20\,000 \text{ cm}^3$

4 Convert these units of volume.

a 2 cm^3 to mm^3

2000 mm^3

e 0.002 km^3 to m^3

$2\,000\,000 \text{ m}^3$

i 0.04 m^3 to mm^3

$40\,000\,000 \text{ mm}^3$

b 2 mm^3 to cm^3

0.002 cm^3

f 0.00002 cm^3 to mm^3

0.02 mm^3

j 0.0016 km^3 to m^3

$1\,600\,000 \text{ m}^3$

c 2 m^3 to cm^3

$2\,000\,000 \text{ cm}^3$

g 0.4 m^3 to cm^3

$400\,000 \text{ cm}^3$

k 0.00016 km^3 to m^3

$160\,000 \text{ m}^3$

d 2 km^3 to m^3

$2\,000\,000\,000 \text{ m}^3$

h 0.04 m^3 to cm^3

$40\,000 \text{ cm}^3$

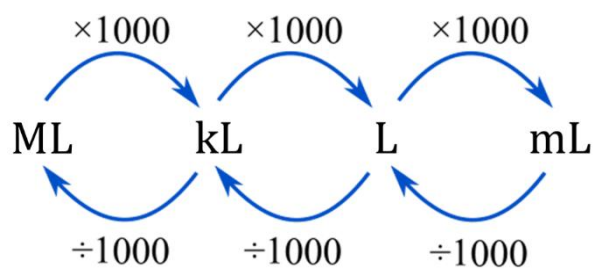
l 0.1 km^3 to m^3

$100\,000\,000 \text{ m}^3$

Units of Capacity

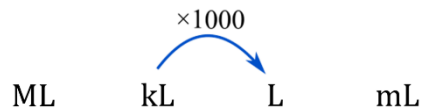
① Converting Units of Capacity

- **Volume** is the amount of space a solid takes up
- **Capacity** is the amount of liquid/gas that an object can hold.
 - A solid block of wood has volume but no capacity. It takes up space but cannot hold anything inside it.
 - A plastic bottle has volume and capacity. It takes up space and can hold liquid or gas.
- Metric units for capacity are millilitres (mL), litres (L), kilolitres (kL), and megalitres (ML).



🔧 Example Convert units of capacity

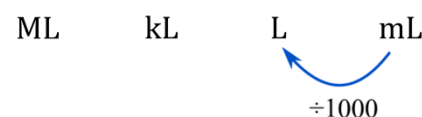
5.7 kL to L



$$5.7 \text{ kL} \times 1000$$

$$= 5700 \text{ L}$$

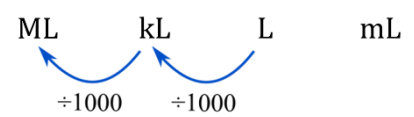
450 mL to L



$$450 \text{ mL} \div 1000$$

$$= 0.45 \text{ L}$$

920 000 L to ML



$$920\,000 \text{ L} \div 1000 \div 1000$$

$$= 0.92 \text{ ML}$$

✎ Guided Practice

Convert these units of capacity.

a 60.42 kL to L

60420 L

b 45 mL to L

0.045 L

c 0.42 kL to mL

420 000 mL

d 0.0304 L to mL

30.4 mL

e 5.43 ML to kL

5430 kL

f 670 400 L to ML

0.6704 ML

Converting Volume and Capacity

$$\begin{aligned}1 \text{ cm}^3 &= 1 \text{ mL} \\1000 \text{ cm}^3 &= 1 \text{ L} \\1 \text{ m}^3 &= 1 \text{ kL}\end{aligned}$$

Example Convert units of capacity

Convert 257 cm³ to mL

257 mL

Convert 32.5 m³ to kL

32.5 kL

Convert 4500 cm³ to L

4.5 L

Guided Practice

Convert these units of capacity.

a 5 cm³ to mL

5 mL

b 2 m³ to kL

2 kL

c 40 000 cm³ to L

40 L

d 2.34 m³ to L

2340 L

e 8920 cm³ to kL

0.00892 kL

f 0.023 cm³ to mL

0.023 mL

Key Ideas

- 1 mL is the amount of fluid/gas that can fit in
- 1 is the amount of fluid/gas that can fit in 1 m³.

1 Convert these units of capacity

a 2000 mL to L 2 L	b 1500 L to kL 1.5 kL	c 900 L to kL 0.9 kL
d 6 L to mL 6000 mL	e 1.25 L to mL 1250 mL	f 0.8 kL to L 800 L
g 1375 mL to L 1.375 L	h 505 L to kL 0.505 kL	i 10 mL to L 0.01 L
j 0.01 kL to L 10 L	k 3.05 kL to L 3050 L	l 5.03 L to mL 5030 mL
m 4070 L to kL 4.07 kL	n 60 mL to L 0.06 L	o 0.07 L to mL 70 mL
p 3.5 ML to L 3 500 000 L	q 890 ML to kL 0.89 kL	r 0.0087 kL to L 8.7 L

2 1 litre = 1000 cm³. Express the following in litres.

a 45 000 cm ³ 45 L	b 800 cm ³ 0.8 L	c 1 m ³ 1000 L
d $\frac{1}{2}$ cm ³ 0.0005 L	e 600 000 mm ³ 0.6 L	

3 Convert these units.

a 4 cm ³ to mL 4 mL	b 2000 cm ³ to L 2 L	c 70 cm ³ to mL 70 mL
d 34 000 m ³ to L 34 000 000 L	e 900 cm ³ to mL 900 mL	f 55 m ³ to L 55 000 L
g 43 m ³ to kL 43 kL	h 3.52 m ³ to L 3520 L	i 79 m ³ to kL 79 kL

4 A pint is an imperial unit for measuring capacity. Five pints is approximately 3 L. Approximately how many mL is one pint?

Around 600 mL

Converting Volume and Capacity

i Converting Volume and Capacity

If you are asked to calculate the capacity of a container:

1. Calculate the volume.
2. Convert to capacity.

Example Calculate capacity of a container

The milk carton has base dimensions 6 cm by 9 cm and height 21 cm.

What is its capacity in litres?

1. Calculate volume

$$\begin{aligned} V &= Ah \\ &= (21)(9)(6) \\ &= 1134 \text{ cm}^3 \end{aligned}$$

2. Convert to capacity

$$\begin{aligned} &= 1134 \text{ mL} \\ &= 1.134 \text{ L} \end{aligned}$$



A cylindrical tank has a radius of 2 m and height of 3 m.

What is its capacity to the nearest whole L?

1. Calculate volume

$$\begin{aligned} V &= \pi r^2 h \\ &= \pi (2)^2 (3) \\ &= 37.69911184 \dots \text{ m}^3 \end{aligned}$$

2. Convert to capacity

$$\begin{aligned} &= 37.69911184 \dots \text{ kL} \\ &= 37\,699 \text{ L} \end{aligned}$$

Guided Practice

Calculate the capacity for these containers.

- a** A plastic takeaway container measuring 12 cm by 17.5 cm by 6.5 cm

1. Volume



2. Capacity

$$1365 \text{ cm}^3$$

$$1.365 \text{ L}$$

- b** A soft drink can with a 3.25 cm radius and 12.3 cm height.

1. Volume

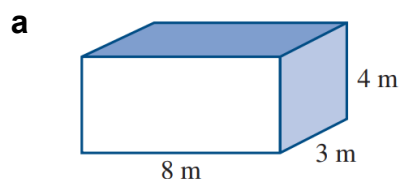


2. Capacity

$$408.15 \text{ cm}^3$$

$$408.15 \text{ mL}$$

- 1 Calculate the volume of these solids, then convert to capacity. Answer in an appropriate unit.

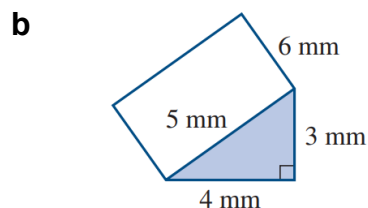


Volume:

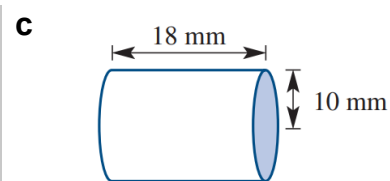
Capacity:

$$96 \text{ m}^3$$

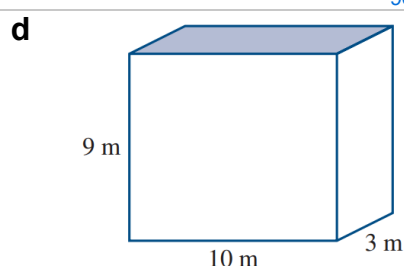
$$96 \text{ kL}$$



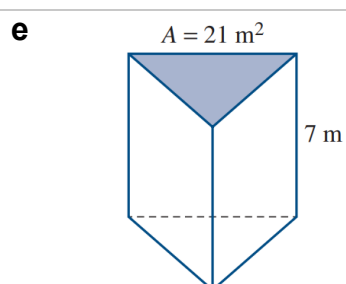
$$36 \text{ mm}^3 = 0.036 \text{ mL}$$



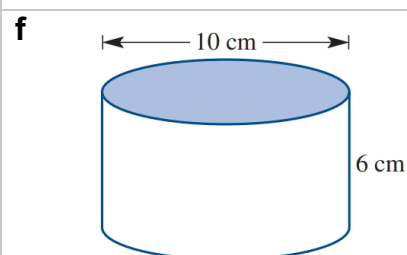
$$5.655 \text{ cm}^3 = 5.655 \text{ mL}$$



$$270 \text{ m}^3 = 270 \text{ kL}$$



$$147 \text{ m}^3 = 147 \text{ kL}$$



$$471.24 \text{ cm}^3 = 471.24 \text{ mL}$$

- 2 A fish tank measures 80 cm by 60 cm by 50 cm.
Find, in litres, the volume of the fish tank.

$$240 \text{ L}$$

- 3 How many litres of fuel can a tanker carry if its tank is cylindrical with a 2 m diameter and is 12 m in length? Round to the nearest litre.

$$37\,700 \text{ L}$$

- 4 Which has a bigger volume and what is the difference in volume to 2 decimal places:
a cube with side length 1 m, or a cylinder with radius 1 m and height 0.5 m?

$$0.57 \text{ m}^3$$

- 5 Samantha is inviting 18 friends to a party.
She estimates that each person will drink 700 mL of soft drink.
- How many litres of soft drink does she need to buy?
 - If each large bottle of soft drink is 2.5 L, how many bottles does she need?

$$19 \times 700 = 13300 \text{ mL} = 13.3 \text{ L}$$

$$13.3 \div 2.5 = 5.32$$

$$\therefore 6 \text{ bottles}$$

- 6 Samira has a 2-litre bottle of cola.
She pours the cola into cylindrical glasses with radius 3 cm and a height of 11 cm.
- What is the capacity of each cylindrical glass?
 - How many full glasses can be filled from the bottle of cola?

$$V = \pi r^2 h = \pi(3)^2(11) = 99\pi \approx 311 \text{ cm}^3$$

$$\text{Number of glasses} = 2000 \div 311 \approx 6.4$$

$$6 \text{ full glasses}$$

- 7 A juice carton measures 8 cm by 8 cm by 15 cm. It is 80% full.
Abby uses the juice to fill some cups. Each cup holds 230 mL.
How many full cups of juice will Abby be able to fill?



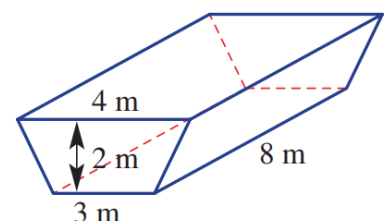
$$\text{Carton volume} = 8 \times 8 \times 15 = 960 \text{ cm}^3$$

$$\text{Juice available} = 960 \times 0.8 = 768 \text{ cm}^3$$

$$\text{Number of cups} = 768 \div 230 = 3.34$$

$$\text{Abby can fill 3 cups}$$

- 8 A swimming pool is a prism with a cross-section that is a trapezium.
- Find the capacity of the pool in litres.
 - The pool is being filled at a rate of 1000 litres per hour.
How long will it take to fill the pool?

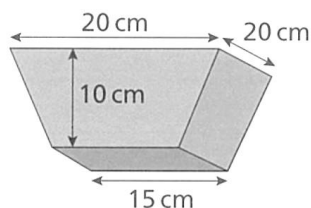


$$\text{Trapezium area} = \left(\frac{2}{2}\right)(3 + 4) = 7 \text{ m}^2$$

$$\begin{aligned} \text{Pool volume} &= Ah = 7 \times 8 = 56 \text{ m}^3 \\ &= 56,000 \text{ L} \end{aligned}$$

$$\text{Time} = 56\,000 \div 1000 = 56 \text{ hours}$$

- 9 Jakub fills an empty flowerpot with soil. The flowerpot is a prism. Its cross-section is a trapezium. Jakub has 2.5 litres of soil. Does Jakub have enough soil to fill the flowerpot?



$$\text{Trapezium area} = \left(\frac{10}{2}\right)(15 + 20) = 175 \text{ cm}^2$$

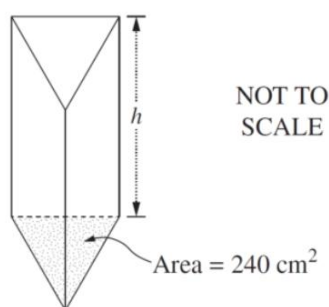
$$\text{Volume needed} = 175 \times 20 = 3500 \text{ cm}^3 = 3.5 \text{ L}$$

Jakub has 2.5 L, so not enough soil.

10 2016 HSC Standard 2 Band 4

A container is in the shape of a triangular prism that has a capacity of 12 litres.

The area of the base is 240 cm^2 . What is the distance, h , between the ends of the container?



$$V = \text{Base area} \times \text{height}$$

$$12 \text{ L} = 240 \text{ cm}^2 \times h$$

$$12000 \text{ cm}^3 = 240h$$

$$h = 50 \text{ cm}$$

11 2010 HSC Standard 2 Band 5

During a flood 1.5 hectares of land was covered by water to a depth of 17 cm.

How many kilolitres of water covered the land? (1 hectare = 10 000 m^2)

(A) 2.55 kL

(B) 2550 kL

(C) 255 000 kL

(D) 2 550 000 kL

$$\text{Area} = 1.5 \text{ ha} \times 10000 \text{ ha/m}^2 = 15000 \text{ m}^2$$

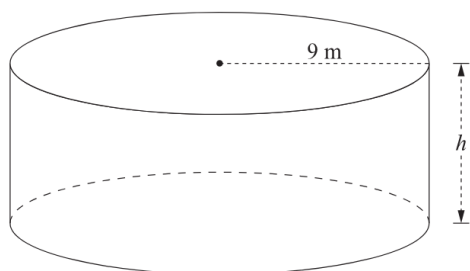
$$\text{Volume} = 15000 \text{ m}^2 \times 0.17 \text{ m} = 2550 \text{ m}^3 = 2550 \text{ kL}$$

B

12 2018 HSC Standard 2 Band 5

A cylindrical water tank has a radius of 9 metres and a capacity of 1.26 megalitres.

What is the height of the water tank? Give your answer in metres, correct to two decimal



$$V = \pi r^2 h$$

$$1.26 \text{ ML} = \pi(9)^2 h$$

$$1\,260\,000 \text{ L} = 81\pi h$$

$$h = \frac{1260000}{81\pi} = 4.95 \text{ m}$$

- 13** Warragamba Dam is filled by runoff from a catchment area of 9050 square kilometres. In a storm, 40 mm of rain fell. What is the increase in the volume of the dam?

$$\begin{aligned}\text{Volume} &= \text{area} \times \text{depth} = 9050 \text{ km}^2 \times 0.04 \text{ m} \\ &= (9050 \times 1\,000\,000) \text{ m}^2 \times 0.04 \text{ m} = 362\,000\,000 \text{ m}^3\end{aligned}$$

- 14** Water is being poured into a fish tank at a rate of 2 L every 10 seconds. The tank is 1.2 m long by 1 m wide by 80 cm high. How long will it take to fill the tank? Give the answer in minutes.

$$\begin{aligned}\text{Tank volume} &= 1.2 \times 1 \times 0.8 = 0.96 \text{ m}^3 = 960 \text{ L} \\ \text{Rate} &= 2 \text{ L per } 10 \text{ seconds} = 0.2 \text{ L/s} = 12 \text{ L/min} \\ \text{Time} &= 960 \div 12 = 80 \text{ minutes}\end{aligned}$$

- 15** A saltwater swimming pool requires 2 kg of salt to be added for every 10 000 litres of water. Joan's pool is a rectangular prism 1.5 metres deep, 5 metres wide and 15 metres long. How much salt will she need to add to her pool? (Hint: $1 \text{ m}^3 = 1000 \text{ L}$)

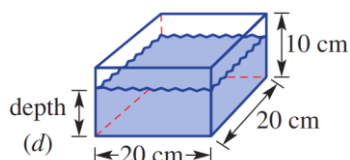
$$\begin{aligned}\text{Pool volume} &= 1.5 \times 5 \times 15 = 112.5 \text{ m}^3 = 112500 \text{ L} \\ \text{Salt needed} &= (112500 \div 10000) \times 2 = 22.5 \text{ kg}\end{aligned}$$

- 16** Each day a factory produces 5000 litres of soft drink in their 13 vats. The soft drink is distributed in cans 13 centimetres high with diameter 6.5 centimetres. How many cans does one vat produce each day?

$$\begin{aligned}\text{Can volume} &= \pi(3.25)^2(13) = 433.7 \text{ cm}^3 \\ \text{Daily production per vat} &= 5000 \text{ L} \div 13 = 384.6 \text{ L} = 384615 \text{ cm}^3 \\ \text{Cans per vat} &= 384615 \div 433.7 = 887 \text{ cans}\end{aligned}$$

- 17 A rectangular container is partially filled with 2.4 litres of water.

a Calculate the base area of the container, in square cm.



$$A = 20 \text{ m} \times 20 \text{ m} = 400 \text{ cm}^2$$

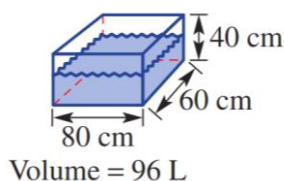
b Calculate the depth of the water using $V = Ah$, where the height is d .

$$2400 \text{ cm}^3 = (400 \text{ cm}^2)h$$

$$h = 6 \text{ cm}$$

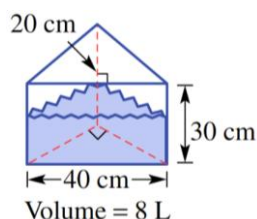
- 18 Use the same method as the previous question to calculate the depth of water in these prisms.

a



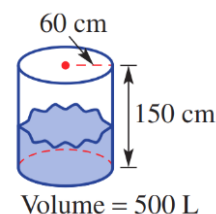
$$20 \text{ cm}$$

b



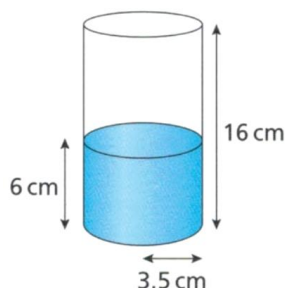
$$20 \text{ cm}$$

c



$$44.2 \text{ cm}$$

- 19 A cylindrical vase is filled with water to the height shown. Sven pours in another 200 ml of water. Calculate the new depth of the water.



First figure out how high 200 mL would be

$$V = Ah$$

$$200 = (\pi(3.5)^2)h$$

$$h = 5.2$$

$$5.2 + 6 = 11.2 \text{ cm}$$

- 20 A pool is 50 metres long, 25 metres wide, 1 m deep at one end and 2 m deep at the other.

a Draw a diagram of the pool. What 3D solid is this?

Trapezoidal prism

b Calculate the capacity of the pool in kilolitres.

$$V = Ah$$

$$V = \frac{50}{2}(1 + 2)(25) = 1875 \text{ m}^3 = 1875 \text{ kL}$$

c If the level of the pool dropped by 75 cm on a hot day, calculate how much water is needed to replace the evaporated water, answer in kilolitres.

$$V = Ah = (25)(50)(0.75) = 937.5 \text{ m}^3$$

$$937.5 \text{ kL}$$